MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY, DISPLAY\PROCESSOR

This specification is approved for use by the US Army Armament, Munitions and Chemical Command and is ava~la151e for use by all Departments and Agencies of the Department of Defense.

1. S COPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the Display/Processor Circuit Card Assembly: 9379229-1, used in the Mortar Btllli.sties Computer (MBC): M23.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926	-	Fire Control Materiel; General specification Governing the Manufacture and Inspection
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-209	Quality Assurance Terms and Definitions
MIL-STD-454	Electronic Equipment, General Requirements for
MIL-STD-1521	Technical Reviews and Audits for Systems, Equipments, and Computer Programs
MIL-STD-45662	Calibration System Requirements

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal.

DRAWINGS

9379229-1	-	Circuit Card Assembly, Display/Processor
9379253		Software, Test Station-Display/ Processor CCA
11785850	·	Computer Set, Ballistics, Mortar: M23

PACKAGING DATA SHEETS

9379229-1	-	Packaging of Circuit Card Assembly, Display\Processod: 9379229-1

(Copies of drawings and packaging data sheets required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Item definition. The display/rocessor circuit card assembly provides the Mortar Ballistic Computer (MBC) with the following:

a. An 8 bit CMOS microprocessor.

b. 2048 8-bit words of CMOS Random Access Memory (RAM).

16 384 8-bit words of Erasable Programmable Read Only Memory (EPROM).

d. Interface circuitry for the keyboard and power supply.

e. A vacuum fluorescent display and interface circuitry.

f. System timing clocks.

The EPROM is located functionally between hexi.decimal addresses 0000 and 3FFF and is board programmable thru' connector P1. The RAM is located functionally between hexideci.mal addresses F800 and FFFF. The major hardware functions for the display\processor circuit card assembly are shown i.n Figure 1 and the connector contacts are listed in Table 1.

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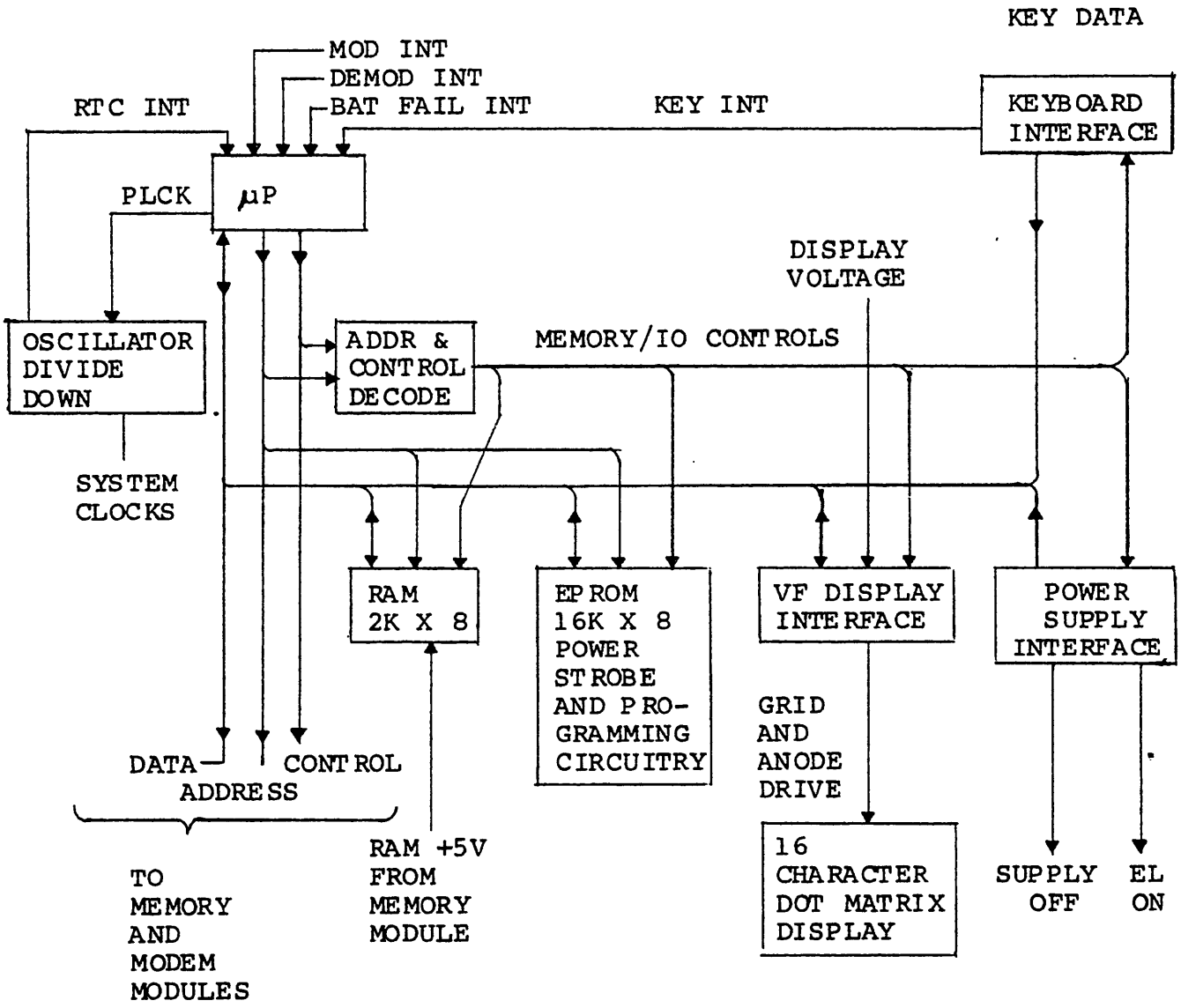


FIGURE 1. Display/processor block diagram.

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TABLE I. Display/processor circuit card assembly connector contacts.

Connector	Pin	Signal Name	Pin	Signal Name
P1	Al	+5V	B1	GND
P1	A2	AB00	B2	DBO
P1	A3	AB0l	B3	DB1
P1	A4	AB02	B4	DB 2
P1	A5	AB03	B5	DB 3
P1	A6	AB04	B6	DB4
P1	A7	AB05	B7	DB 5
P1	A8	AB06	B8	DB6
P1	A9	AB07	B9	DB7
P1	Al	AB08	B10	RD*
P1	All	AB09	Bll	WT*
P1	A12	ABlo	B12	10/M*
P1	A13	ABll	B13	SO
P1	A14	AB12	B14	S1
Pl	A15	AB13	B15	RESET*
P1	A16	AB14	B16	LE
P1	A17	AB15	B17	ALE
P1	A18	+5V	B18	GND
P1	A19	PCLK	B19	PCLK
P1	A20	KEYINT*	B20	KEYINT*
P1	A21	RTCINT*	B21	RTCINT*
P1	A23	DEMODINT*	B 23	RAM+5V
P1	A24	PSFAULT*	B24	WAIT*
Pl	A25	BREQ*	B25	BACK*
P1	A26	LBO*	B26	HBO*
P1	A27	LB1*	B27	HB1
P1	A28	LB2*	B28	HB2*
PI	A29	LB3*	B29	HB3*
P1	A3 0	LB4*	B30	HB4*
P1	A31	LB5*	B31	HB5*
P1	A32	LB6*	B32	HB6*
P1	A33	+5V	B33	GND
P1	A34	Spare	B34	614.4KHz
P1	A35	Spare	B35	5Hz
P1	A36	SENSE1	B36	2HZ
P1	A37	SENSE2	B37	lHz
P1	A38	p ~ *	B38	2SEC
P1	A39	VVP	B39	3 SEC
P1	A40	WAIT*	B40	6SEC
P1	A41	OFF	B41	LEDDC
P1	A42	ELON	B42	DSPON*
P1	A43	ANDAT	B43	ANDAT
P1	A44	ANCLK	B44	ANCLK
P1	A45	GRDEN	B45	GRDEN
P1	A46	GRDCLK	B46	GRDCLK

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TABLE 1. Display\processor circuit card assembly connector contacts.-Continued

P1	A47	EOF	B47	EOF
P1	A48	320Hz	B48	320Hz
P1	A49	CNTRST	B49	+35V
P1	A50		B50	GND

3.2 Performance.

3.2.1 General. The performance of the display\processor circuit card assembly shall be in accordance with the following requirements.

3.2.1.1 Functional. The display/processor circuit card assembly shall be attached to the Computer Automation CA8200 tester or equivalent, using the connections as specified in Table IV and pass the test patterns induced by Test Station Software-Display/Processor CCA: 9379253.

3.2.1.2 Memory read cycle. A memory read cycle from either RAM or EPROM shall be accomplished by applying the signals specified in Figure 2 and Table I.

3.2.1.3 RAM write cycle. A RAM write cycle shall be accomplished by applying the signals specified in Figure 3 and Table I.

3.2.1.4 EPROM program cycle. An EPROM program cycle shall be accomplished by applying the signals specified in Figure 4 and Table I.

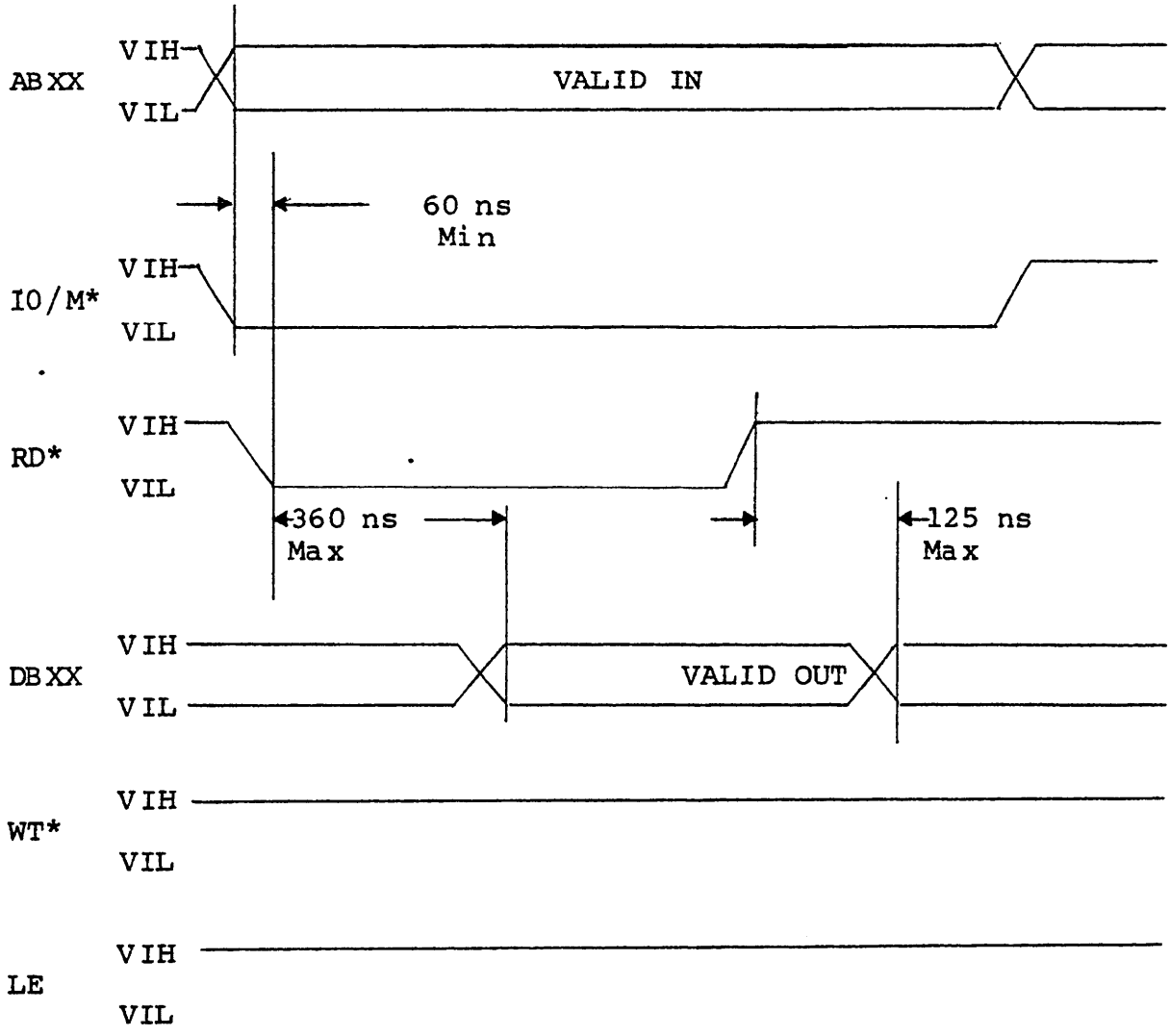
3.2.1.5 EPROM erasure. The EPROM shall be considered to be erased when hexadecimal memory locations (0000) to (3FFF) contain data (FF).

3.2.2 Physical characteristics. The physical characteristics shall be as prescribed by the drawings, specifications, and QAP's.

3.2.3 Reliability. When specified in the contract, a special sample shall be subjected to reliability testing. The manufacturing and production techniques emloyed in the fabrication of the display/processor circuit card assembly shall not degrade the inherent reliability. The inherent reliability shall not be less than 59,385 hours Mean Time Between Failures (MTBF). (See 4 and 6)

3.3 Environmental. The display/processor circuit card assembly shall meet the requirements specified herein unless otherwise specified.

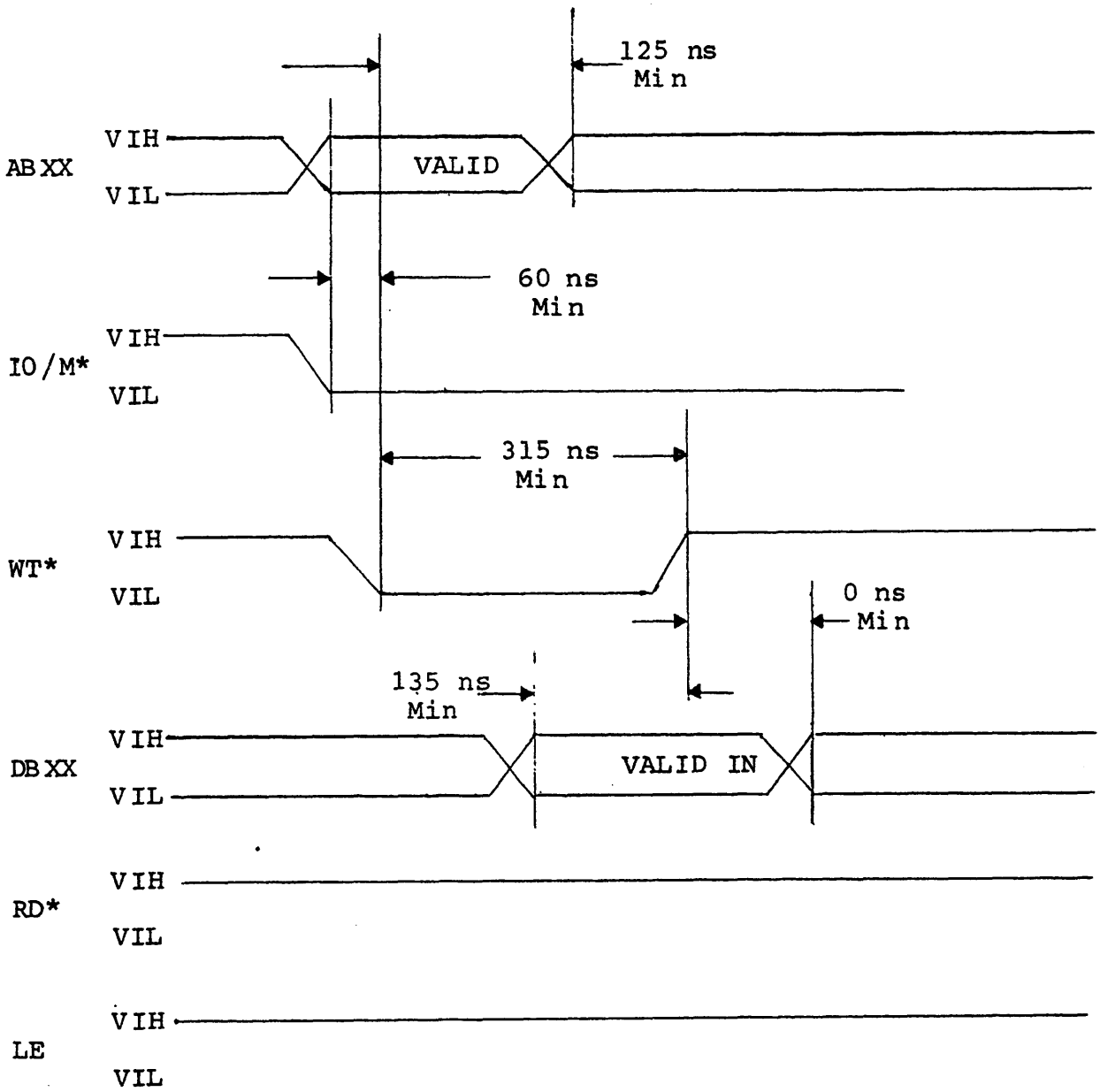
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VIL AND VOL ARE 0.5 ± 0.5 VOLTS DC
VIH AND VOH ARE 4.6 ± 0.7 VOLTS DC

FIGURE 2. Memory read cycle.

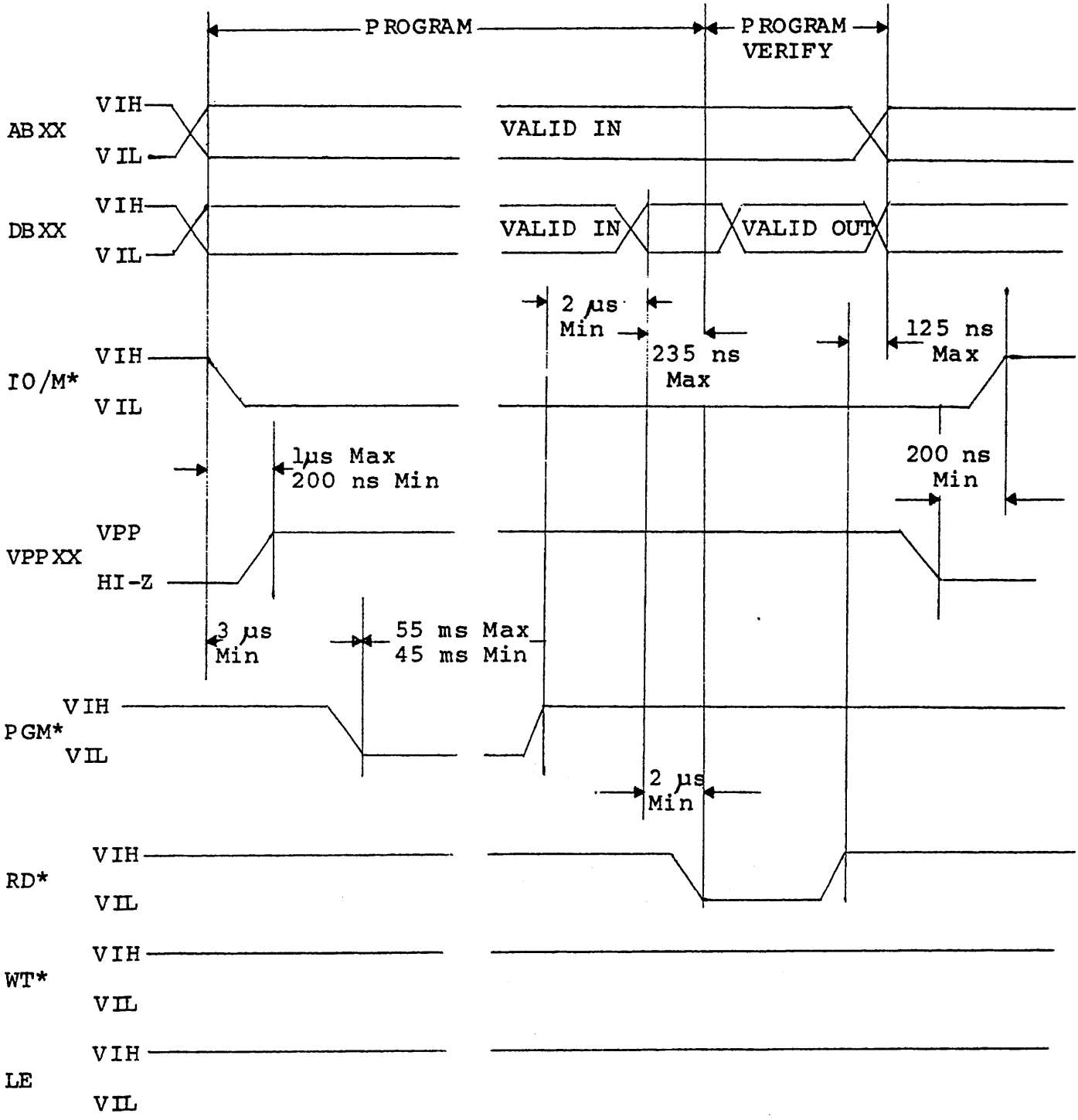
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VIL AND VOL ARE 0.5 ± 0.5 VOLTS DC
VIH AND VOH ARE 4.6 ± 0.7 VOLTS DC

FIGURE 3. Ram write cycle.

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VIL AND VOL ARE 0.5 ± 0.5 VOLTS DC
VIH AND VOH ARE 4.6 ± 0.7 VOLTS DC

FIGURE 4. EPROM program cycle.

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3.3.1 Room ambient temperature. The requirements of this specification shall be met with the display/processor assembly thermally stabilized at room ambient temperature of +16°C to +32°C (+60°F to +90°F), unless otherwise specified in the requirements paragraph.

3.3.2 Environmental stress screen (ESS) temperature. The display/processor circuit card assembly shall meet the requirements of 3.2.1.1 and 3.2.1.5-inclusive, after having been exposed to a minimum of ten temperature cycles at temperatures of -57°C and +71°C (-71°F and +160°F).

3.3.3 Environmental stress screen (ESS) vibration. The display/processor circuit card assembly shall meet the requirements of 3.2.1.1 and 3.2.1.5 inclusive, after having been" subjected to a random or pseudorandom vibration which includes frequency content from 100 Hz to 1000 Hz and UP to a 3q RMS in two axis for a Period of 10 minutes plus or minus is seconds.

3.4 Fabrication. The display/processor circuit card assembly shall be manufactured in accordance with drawing 9379229-1.

3.5 First article. When specified in the contract, a sample shall be subjected to first article inspection (See 4 and 6).

3.6 Workmanship. The workmanship requirements of MIL-F-13926 and MIL-STD-454 shall be met. All components shall be fitted properly and firmly in their respective positions. All electrical connections shall be mechanically secure and electrically sound. There shall be no evidence of corrosion burrs, roughness, sharp edges or other defects which could adversely affect its use or suitability for the purpose intended.

3.7 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 and MIL-STD-454 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval test shall consist of three (3) display/circuit card assemblies plus three (3) each of all items covered by QAP. The three display/circuit card assemblies shall be tested as specified herein for"all the requirements of this specification. The items covered by QAP shall be tested, as specified thereon, for all characteristics delineated on the QAP. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for-all the requirements-of the contract. The sample and a record of this inspection, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractor's inspections.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting officer, inspection lot size, lot formation, and presentation of lots shall be in accordance with "submission of Product" provision of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Quality Assurance Provisions (QAP's) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

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4.4 Sampling.

4.4.1 Environmental\performance. The display\processor circuit board assembly shall be inspected to the requirements in Table II.

TABLE II. Classification of defects.

No.	Characteristics	Requirement	Test Procedure
100% Inspection			
1.	Functional test	3.2.1.1	4.6.3
2.	Memory read cycle	3.2.1.2	4.6.4
3.	RAM write cycle	3.2.1.3	4.6.5
4.	EPROM program cycle	3.2.1.4	4.6.6
50	EPROM erasure	3.2.1.5	4.6.7
6.	ESS temperature	3.3.2	4.6.1
7.	ESS vibration	3.3.3	4.6.2
MAJOR: AOL 1.0% Defective			
101.	Workmanship	3.6	4.6.8

4.4.2 Special sampling. One display/processor circuit board assembly shall be selected at “random as a special sample from each lot or each 100 produced which ever occurs first. The sample shall meet the requirements and tests in Table III.

TABLE III. Classification of defects.

No.	Characteristics	Requirement	Test Procedure
201.	Fabrication	3.4	Applicable Drawing/ Visual
202.	Materials	3.7	Applicable Drawing/ Visual

NOTE : The tests in Table III shall be conducted at room ambient temperature (as specified in 3.3.1).

4.4.3 Failure of sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and resubmitted items have been approved in accordance with “Acceptance and Rejection” as specified in MIL-STD-105.

4.5 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

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4.5.1 Government furnished inspection equipment. Where the contract provides for Government furnished test equirpment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. All inspection equipment specified by drawing number in specification or QAP frming a part of the contract shall be supplied by the contractor in accordance with technical data included in the Technical Data Package List (TDPL).

4.5.2.2 Contractor design. In the absence of Government inspection equirement design, the contractor shall design and supply inspection equipment compatible with the “Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in “Examination” and "Test FAcilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Temperature testing. The volume of the test chamber shall conform to the test tactilities requirements of MIL-F-13926. The air temperature at the control sensor shall be in accordance with the specified requirements within plus or minus 2.5°F (1.4°C). The temperature gradient across the cross-sectional area occupied by the test item shall not exceed 0.5°F per foot in any direction. The heat source of the test chamber shall be so located that radient heat, if used, will not fall directly on the test item. Thermocouples-or equivalent temperature sensors utilized to determine or control the specified chamber temperature shall be centrally located within the test chamber within the supply airstream, or in the return airstream, whichever provides the specified test condition at the item under test and shall be baffled or otherwise protected against radiation effects.

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4.5.2.3,2 Fabrication, setup and calibration Design and fabrication of test equipment shall be such that it incorporates the features of rigidity, stability, and ease of maintenance. Setup and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment. Operating and calibration instruction shall be prepared and submitted to the procuring activity.

4.6 Test methods and procedures. Unless otherwise specified in the contract, the following procedures shall be utilized to conduct specified test. If the contractor chooses to use alternate test procedures, a detailed description of the set-up and procedure shall be submitted to the government procuring agency for approval prior to use.

4.6.1 ESS temperature. Testing equipment utilized in this test shall conform to the requirements of 4.5.2.3.1. The display/processor circuit card assembly shall be placed in the test chamber and the temperature of the chamber reduced 10°C (18°F) per minute plus or minus l°C (1.8°F) from room ambient temperature (as specified in 3.3.1) to -57°C (-71°F). The assembly shall be maintained at this temperature for a minimum of 20 minutes. Upon completion of the 20 minute exposure, the temperature of the chamber shall be raised 10°C (18°F) per minute plus or minus 1°C (1.8°F) to +71°F (+160°C). The assembly shall remain at this temperature for a minimum of 20 minute Upon completion of the 20 minute exposure the temperature of the chamber shall be reduced 10°C (18°F) per minute plus or minus l°C (1.8°F) to room ambient temperature (as specified in 3.3.1). The above procedure shall be conducted a minimum of ten times. Upon completion of the cycles the assembly shall then be removed from the chamber and subjected to the test in 4.6.3. It shall meet the requirements of 3.2.1.1.

4.6.2 ESS vibration. Conduct the vibration test with the display/processor circuit card assembly positioned on an appropriate adapter affixed to a vibrating machine that is capable of providing the amplitudes and frequencies specified in 3.3.3. Unless otherwise specified, the test equipment shall be in accordance with the “Test Facilities” requirements of MIL-F-13926. The applied motion shall be in two separate planes for the time period specified. Upon completion of the vibration sweep the assembly shall be removed from the vibration adapter and shall be subjected to the test in 4.6.3. It shall meet the requirements of 3.2.1.1.

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4.6.3 Functional. This test is to be conducted utili.zng the Computer Automation CA8200 or equivalent. Use of an equivalent computer must have prior approval of the procuring activity. The display\processor circuit card assembly shall be attached to the CA8200 utilizing appropriate fixtures and\or cables and connected as specified in Table I_V. 'The assembly shall be tested in accordance with the test patterns contained on the-Test Station Software-Display\Processor CCA: 9379253 provided by the procuring activity. The assembly shall meet the requirements of 3.2.1.1.

TABLE IV. Display/processor circuit card assembly test fixture connector contacts.

Display/Processor		Connect to	CA8200
Connector	Pin		Pin
P1	Al		V0+
P1	A2		322
P1	A3		323
P1	A4		324
P1	A5		325
P1	A6		326
P1	A7		327
Pl	A8		328
P1	A9		329
P1	A10		330
Pl	All		331
P1	A12		332
P1	A13		333
P1	A14		334
P1	A15		335
P1	A16		336
P1	A17		337
P1	A18		V0+
P1	A19		339
P1	A20		340
P1	A21		341
P1	A22		342
P1	A23		343
P1	A24		344
Pl	A25		345
P1	A26		346
P1	A27		347
P1	A28		348
P1	A29		349
P1	A30		350
P1	A31		351
P1	A32		352
P1	A33		V1+
PI	A34		354

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TABLE IV. Display/processor circuit card assembly test fixture connector contacts.-Continued

Display/Processor		Connect to	CA8200
Connector	Pin		Pin
P1	A35		355
P1	A36		356
P1	A37		357
P1	A38		358
P1	A39		359
P1	A40		360
P1	A41		361
P1	A42		362
P1	A43		363
P1	A44		364
P1	A45		365
P1	A46		366
P1	A47		367
P1	A48		368
P1	A49		369
P1	A50		370
P1	B1		GND
P1	B2		372
P1	B3		373
P1	B4		374
P1	B5		375
P1	B6		376
P1	B7		377
P1	B8		378
P1	E9		379
P1	B1O		380
P1	Bll		381
P1	B12		382
P1	B13		383
Pl	B14		384
P1	B15		385
P1	B16		386
P1	B17		387
P1	B18		GND
P1	B19		389
P1	B20		390
P1	B21		391
P1	B22		392
PI	B23		V3+
P1	B24		394
P1	B25		395
P1	B26		396
P1	B27		397
P1	B28		398

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TABLE IV. Display/processor circuit card assembly test fixture connector contacts.-Continued

Display/Processor		Connect to	CA8200
Connector	Pin		Pin
P1	B29		399
PI	B30		400
P1	B31		401
P1	B32		402
P1	B33		GND
P1	B34		404
P1	B35		405
P1	B36		406
P1	B37		407
P1	B38		408
P1	B39		409
P1	B40		410
P1	B41		411
P1	B42		412
P1	B43		413
P1	B44		414
P1	B45		415
P1	B46		416
PI	B47		417
P1	B48		418
P1	B49		Vz+
P1	B50		GND
TP	1		421
TP	2		422
TP	3		423
TP	4		424
TP	5		425
TP	6		426
TP	7		427
TP	8		428
TP	9		429
TP	10		430
TP	11		431
TP	12		432
TP	13		433
TP	14		434
TP	15		435
TP	16		436
TP	17		437
TP	18		438
TP	19		439
TP	20		440
TP	21		441
TP	22		442

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TABLE IV. Display/processor circuit card assembly test fixture connector contacts.-Continued

Display/Processor Connector	Pin	Connect to	CA8200 Pin
TP	23		443
TP	24		444
TP	25		445
TP	26		446
TP	27		447
TP	28		448
TP	29		449
TP	30		450
TP	31		451
TP	32		452
TP	33		453
TP	34		454
TP	35		455
TP	36		456
TP	37		457
TP	38		458
TP	39		459
TP	40		460
TP	41		461
TP	42		462
TP	43		463
TP	44		464
TP	45		465
TP	46		466
TP	47		467
TP	48		468
TP	49		469
TP	50		470
TP	51		471
TP	52		472
TP	53		473

4.6.4 Memory read cycle. Tests 4.6.5 and 4.6.6 shall be perfotmed by first connecting the following connector contacts together and then applying a 5.3 plus or minus .1 volt dc power supply to P1-A1 (+5-V) with respect to P1-B18 (GND). The display\processor assembly shall be initilized for memory tests by connecting P1-A25 (BREQ*) to GND and then momentarily connecting P1-B15 (RESET*) to GND.

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Memory Test Connections

P1-A1 to P1-A33	P1-A45 to P1-B45
P1-A1 to P1-B23	P1-A46 to P1-B46
P1-A19 to P1-B19	P1-A47 to P1-B47
P1-A43 to P1-B43	P1-A48 to P1-B48
P1-A44 to P1-B44	P1-A25 to P1-B1

4.6.5 RAM write cycle.

Write to RAM with hexadecimal data (01) using the RAM write cycle specified in 3.2.1.3.

b. -Verify the data written to RAM is correct by reading data from RAM using the memory read cycle described in 3.2.1.2.

c. Rotate to the right by 1 bit the data in the memory location being tested and reverify the data for each rotation.

d. Data shall be written as in a. thru c. to the full range of RAM from hexadecimal address (F800) to address (FFFF).

e. Write to the full range of RAM from hexadecimal address (F800) to address (FFFF) with hexadecimal data (00) using the RAM write cycle specified in 3.2.1.3.

f. Write to the RAM location being tested with hexadecimal data (FF) using the RAM write cycle specified in 3.2.1.3.

g. Verify that all RAM addresses that differ from the test address by 1 bit contain hexadecimal data (00) by using the memory read cycle described in 3.2.1.2.

h. Data shall be written as in e. thru g. to the full range of RAM from hexcidecimal address (F800) to address (FFFF).

4.6.6 EPROM program cycle. An EPROM program cycle shall be accomplished by applying the signals specified in Figure 3.4 and Table 1. P1-A39 shall have the input parameters as specified on the EPROM's respective drawing.

4.6.7 EPROM erasure. The EPROM shall be erased by exposing the transparent window to high intensity ultraviolet light. A light source with a wavelength of 2537 Angstroms yielding. a minimum integrated dosage of 15 w-soO\cm(2) is required. An ultraviolet light source with a power of 12,000 microwatts\cm(2) placed within 1 inch of the window will erase the EPROM in 20 minutes. The EPROM shall be erased when requirement 3.2.1.5 is met.

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4.6.8 Workmanship. “The display/processor circuit card assembly shall be subjected to a visual and tactile examination for evidence of faulty workmanship. The assembly shall conform to the requirements of 3.6.

4.7 Reliability testing. ‘The requirement for reliability test approval and the responsibility (Government or contractor) for reliability testing shall be as specified in the contract. The sample for reliability test shall consist of 3 display/processor circuit card assemblies. The samples shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as will be used in the production of the balance of the contract quantity. All parts and materials including packaging and packing, shall be obtained from the same source of supply as will be used in the production of the balance of the contract quantity.

4.7.1 Government testing. When the Government is responsible for conducting reliability tests, the contractor prior to submitting the sample to the Government, shall inspect the sample to assure that it conforms to all requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for all materials.

4.7.2 Contractor testing. When the contractor is responsible for conducting reliability tests, the sample shall be inspected and tested by the contractor for all the requirements of the contract. The sample and a record of these inspections and tests, including certificates of conformance for all materials, shall be submitted to the Government for approval. The Government reserves the right to witness any or all of the contractors reliability tests and inspections.

5. PREPARATION FOR DELIVERY

5.1 Delivery requirements. Handling, packaging, packing, shipping, and marking of the display/processor circuit card assembly, 9379229-1 shall be in accordance with Packaging Data Sheet 9379229-1. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The display/processor circuit card assembly described herein is intended for use in the MBC. See drawing 11785850.

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6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Provisions for first article testing.

c. Provisions for reliability testing.

d. Applicable Packaging Data Sheet.

Applicable levels of preservation, packing, and marking (see 5.1).

f. Applicable stock numbers.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1220-A299)

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-70491 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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